

CONTROL THEORY-1

CONTROL THEORY-1 ► General ► الامتحان الالمقي

Question 3

Not answered

Scored out of 2.00

Flag question

The following system $T(s) = \frac{2s+1}{(s^2+25)}$ is

Select one:

- ☐ a. underdamped
- ☐ b. overdamped
- ☐ c. critical damped
- ☒ d. undamped

CONTROL THEORY-1

CONTROL THEORY-1 ► General ► الامتحان التصفي

Question 18

Not yet answered

Marked out of 1.00

Flag question

Poles of the input function determine the form of the transfer function.

Select one:

☐ True

☒ False

CONTROL THEORY-1

CONTROL THEORY-1 ► General ► الامتحان التمهيدى

Question 4

Not yet answered

Marked out of 2.00

Flag question

Given the following transfer function

$$G(s) = \frac{144}{s^2 + 6s + 12}$$

The exponential decay frequency equals to

Select one:

- ☐ a. 0.25
- ☐ b. 12
- ☐ c. 144
- ☐ d. 6

معلمة التخميد

CONTROL THEORY-1

CONTROL THEORY-1 ► General ► الامتحان الصفّي

16

answered

out of 2.00

question

The transfer function of the system is $G(s) = 100/(s+1)(s+100)$. For a unit step input to the system the approximate settling time (in sec) for 2% criterion is:

Select one:

- ☐ a. 1
- ☐ b. 100
- ☐ c. 4
- ☐ d. 0.04

Next page

Question 13

Not yet answered

Marked out of 2.00

Flag question

The following system $T(S) = \frac{10}{(s+7)(s+3)}$ is

Select one:

- ☐ a. undamped
- ☒ b. overdamped
- ☐ c. underdamped
- ☐ d. critical damped

CONTROL THEORY-1

CONTROL THEORY-1 ► General ► الامتحان اللفظي

A pole on the real axis generates an exponential response

Select one:

- ☒ True
- ☐ False

(Not oscillated)

CONTROL THEORY-1

3444-CONTROL THEORY-1 ► General ► الامتحان النصفى

Question 14

Not yet answered

Marked out of 2.00

Flag question

If $c(t) = 3 + 5e^{-4t}$, the forced response is

Answer:

3

quiz/attempt.php?attempt=150190

answered

out of 2.00

g question

Given the following differential equation

$$\frac{d^2 y}{dt^2} + 3 \frac{dy}{dt} + 2y = u(t)$$

$Y(s) =$

Select one:

☐ a.

$$\frac{-(s^2 + s - 1)}{s(s^2 + 3s + 2)}$$

☐ b.

$$\frac{(s^2 + s - 1)}{(s^2 + 3s - 2)}$$

☒ c.

$$\frac{1}{s(s^2 + 3s + 2)}$$

☐ d.



tempt.php?attempt=150190&page=1

CONTROL THEORY-1

CONTROL THEORY-1 ► General ► الامتحان التصفي

red
2.00
ion

The poles of the following transfer function $G(s) = \frac{2s+1}{s^2+4s+3}$ are

Select one:

- ☐ a. 1 , -3
- ☐ b. 1 , 3
- ☒ c. -3 , -1
- ☐ d. 3 , -1

Not yet answered

Marked out of 2.00

 Flag question

Integral control of first order system

Select one:

- ☐ a. decrease maximum overshoot
- ☐ b. reduce the steady state error
- ☐ c. reduce settling time
- ☒ d. eliminate steady state error

Next page

Not yet answered

Marked out of 2.00

 Flag question

The transfer function of first order system is $C(s)/R(s) = 1/(4s+2)$. assume unit step input, the time constant is

Select one:

☐ a. 0.5

☐ b. 4

☐ c. 1

☒ d. 2

Next page

Not yet answered

Marked out of 2.00

 Flag question

The transfer function of second order system is $C(s)/R(s) = 1/(s^2 + 2s + 4)$. assume unit step input, the damping constant is

Select one:

☐ a. 0.5

☐ b. 4

☒ c. 1

☐ d. 2

Next page

Not yet answered

Marked out of 2.00

 Flag question

In force-current analogy, velocity is analogous to

Select one:

- ☐ a. charge
- ☐ b. inductance

☒ c. Voltage

Voltage

- ☐ d. Capacitance

Next page

Not yet answered

Marked out of 2.00

 Flag question

At which frequency does the magnitude of the system becomes zero dB

Select one:

- ☐ a. Cut-off frequency
- ☐ b. Resonant frequency
- ☐ c. Phase crossover frequency
- ☒ d. Gain crossover frequency

Next page

Not yet answered

Marked out of 2.00

 Flag question

The proportional controller is mainly used to

Select one:

- ☐ a. reduce settling time
- ☐ b. increase maximum overshoot
- ☐ c. destabilize the open loop system
- ☒ d. reduce the steady state error

Next page

Not yet answered

Marked out of 2.00

 Flag question

The slope of logarithm of magnitude vs. frequency of lead first order factor is

Select one:

- ☐ a. +20dB/ dec at all frequencies
- ☐ b. Zero dB/dec at very low frequencies and +20dB/ dec at very high frequencies
- ☐ c. Zero dB at all frequencies
- ☒ d. -20dB/ dec at all frequencies

Next page

Not yet answered

Marked out of 2.00

🚩 Flag question

Consider the system represented by the equation given below. What would be the total phase value at $\omega = 0$

$$G(s)H(s) = 200/[s^3 (s + 3) (s + 6) (s + 10)].$$

Select one:

- ☐ a. -90
- ☐ b. -180
- ☐ c. zero
- ☐ d. -270

?

Next page

Not yet answered

Marked out of 2.00

 Flag question

In closed loop transfer function, the nominator is

Select one:

- ☒ a. forward transfer function
- ☐ b. forward and open loop transfer function

Next page

Not yet answered

Marked out of 2.00

 Flag question

The result of proportional control of first order system is

Select one:

- ☐ a. Increase the time constant
- ☐ b. The system always unstable
- ☒ c. Decrease the time constant
- ☐ d. Increase the maximum overshoot

Next page

Not yet answered

Marked out of 2.00

Flag question

The root locus is the locus of the change of thetraced out in the s-plane.

Select one:

- ☐ a. As the input of the system is changed
- ☐ b. As the sensitivity is changed
- ☒ c. As a system parameter is changed
- ☐ d. As the output of the system is changed

Next page

Not yet answered

Marked out of 2.00

 Flag question

The unit ramp response of standard first order system is

Select one:

- ☐ a. $(1-e^{-t})/T$
- ☐ b. $+e^{-t}/T$
- ☒ c. none
- ☐ d. $-e^{-t}/T$

Next page

Not yet answered

Marked out of 2.00

🚩 Flag question

When the function $F(s)$ has distinct poles, the type of the function $f(t)$ is

Select one:

- ☐ a. undamped
- ☐ b. damped ramp
- ☐ c. critically damped sinusoidal
- ☒ d. Exponential

Next page

Not yet answered

Marked out of 2.00

 Flag question

The characteristic equation of a closed loop system is $S^3+s^2+2s+1=0$, this system is

Select one:

- ☐ a. critically stable
- ☐ b. stable
- ☐ c. conditionally stable
- ☒ d. unstable

Next page

January - 6 February 31

FINAL EXAM 1.02.2021

سؤال 1

غير مجاب عليه بعد

الدرجة من 2.00

علم هذا السؤال

The transfer function of second order system is $C(s)/R(s) = 1/(s^2 + 2s + 4)$. assume unit step input, the damping constant is

اختر أحد الخيارات

a. 2 ☐b. 0.5 ☐c. 1 ☒d. 4 ☐

الصفحة التالية

تنقل الاختبار

8	7	6	5	4	3	2	1
16	15	14	13	12	11	10	9

When the function $F(s)$ has multiple poles, the type of the function $f(t)$ is

Select one:

- ☒ a. Exponential
- ☐ b. undamped
- ☐ c. damped ramp
- ☐ d. critically damped sinusoidal

CONTROL THEORY-3

30103444-CONTROL THEORY-3 ▶ 31 January - 6 February ▶ FINAL EXAM 1.02.2021

Question 1

Not yet answered

Marked out of 2.00

Flag question

For standard translation mechanical system, The output signal to the system is

Select one.

- ☐ a. applied force
- ☐ b. acceleration
- ☒ c. displacement
- ☐ d. velocity

You are logged in as 21715018510 (Log out)

2021/20204-CONTROL THEORY-3

View profile

CONTROL THEORY-3-202

January - 6 February 31 ◀ CONTROL THEORY-3-2021/2020-1-30103444 ▶

The corner frequency of a standard lag first order factor is

اختر أحد الخيارات

a. T ☒

b. $4T$ ☐

c. 1 ☐

d. $3T$ ☐

تم دخولك بصفة 31415018080 أحمد تيسى حسين هلال (د)
ROL THEORY-3-2021/2020-1-30103444



THEORY-3

THEORY-3 ► 31 January - 6 February ► FINAL EXAM 1.02 .2021

The transfer function of second order system is $C(s)/R(s) = 1/(s^2 + 2s + 4)$. assume unit step input, the damping ratio

Select one:

- ☐ a. 4
- ☒ b. 0.5
- ☐ c. 2
- ☐ d. 1

CONTROL THEORY-3

44-CONTROL THEORY-3 ▶ 31 January - 6 February ▶ FINAL EXAM 1.02.2021

First method of Ziegler Nichols rule used when the time response of the closed loop system

Select one:

- ☐ a. over damped
- ☐ b. undamped
- ☒ c. none
- ☐ d. under damped



علم هذا السؤال

The transfer function of second order system is $C(s)/R(s) = 1/(s^2 + 2s + 4)$. assume unit step input, the undamped natural frequency is

اختر أحد الخيارات

a. 2 ☒b. 3 ☐c. 4 ☐d. 1 ☐

٧٧

الصفحة التالية

+ x

Google | ترجمة - بحث

Root Locus - Control Systems Q

تحميل: سنوات كنترول - we

s2.ebalqa.courses/nfetmoodle/mod/quiz/attempt.php?attempt

Google

30

CONTROL THEORY-3-202

FINAL EXAM 1.02.2021 < January - 6 February 31 < CONTROL THEORY-3-2021/2020-1-30103444 <

Consider the loop transfer function $K(s+6)/[(s+3)(s+5)]$. In the root locus diagram centroid will be located at σ .

اختر أحد الخيارات

a. -1 ☐

b. -2 ☒

c. -4 ☐

d. -3 ☐

تم بحسنه 31415018080 حمد محمد حسن (مدير المخرج)
CONTROL THEORY-3-2021/2020-1-30103444



THEORY-1

الصفحة الرئيسية

First Semester 2021/2020

CONTROL THEORY-1-2021/2020-1-30103444

January - 6 February 31

FINAL EXAM 1.02.2021

سؤال 6

غير مجاب عليه بعد

الدرجة من 2.00

علم هذا السؤال

the signal flow graph was used to obtain

اختر أحد الخيارات

- a. closed loop transfer function ☐
- b. open loop transfer function ☐
- c. feed back transfer function ☐
- d. forward transfer function ☒

الصفحة التال

تنقل الاختبار

3	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

The phase shift of a second order lag factor when the frequency changed from zero to infinity is

Select one:

- ☒ a. Changed from zero degrees up to -180 degrees
- ☐ b. Changed from zero degrees up to $+180$ degrees
- ☐ c. Changed from zero degrees up to $+90$ degrees
- ☐ d. Changed from zero degrees up to -90 degrees

2020-1-30103444-CONTROL THEORY-3

First Semester 2021/2020 > 2021/2020-1-30103444-CONTROL THEORY-3 > 31 January - 6 February > EXAM 1 02 2021

NAVIGATION

3	4	5	6	7
10	11	12	13	14
17	18	19	20	21
24	25			

Question 6

Not yet answered

Marked out of 2.00

Flag question

If the damping ratio of second order system is unity, the response is

Select one:

- ☐ a. over damped
- ☐ b. undamped
- ☐ c. critically damped
- ☐ d. under damped

مسئله واضحة

You are logged in as 31915018070 (Log out)
2021/2020-1-30103444-CONTROL THEORY-3

January - 6 Feb
FINAL EXAM 1.0

ب عليه بعد

2.00 من

علم هذا السؤال

For root loci which of the following are the starting points

اختر أحد الخيارات

- a. Closed loop poles ☐
- b. Open loop zeros ☐
- c. Closed loop zeros ☐
- d. Open loop poles ☒

While moving a branch point after the block, which among the following should be added

Select one:

- ☐ a. summing point
- ☐ b. Block of the same transfer function
- ☒ c. block of reciprocal transfer function
- ☐ d. node



L THEORY-3

L THEORY-3 ► 31 January - 6 February ► FINAL EXAM 1.02 .2021

The phase shift of a second order lag factor when the frequency changed from zero to infinity is

Select one:

- ☐ a. Changed from zero degrees up to +180 degrees
- ☒ b. Changed from zero degrees up to -180 degrees
- ☐ c. Changed from zero degrees up to =90 degrees
- ☐ d. Changed from zero degrees up to -90 degrees

103444-CONTROL THEORY-3

2020 ▶ 2021/2020-1-30103444-CONTROL THEORY-3 ▶ 31 January - 6 February ▶ FINAL EXAM 1.02 2021

Question 6

Not yet answered

Marked out of 2.00

Flag question

If the damping ratio of second order system is unity, the time response is

Select one:

- ☐ a. over damped
- ☐ b. undamped
- ☒ c. critically damped
- ☐ d. under damped

the open loop transfer function is

$$G(s)H(s) = \frac{s + 1}{s(2 + 2)(s + 3)}$$

, The

Nyquist plot of this system terminates at

اختر أحد الخيارات

- ☒ a. zero magnitude and -180 degrees phase angle
- ☐ b. -180 degrees phase angle
- ☐ c. infinity magnitude and -180 degrees phase angle
- ☐ d. zero magnitude and -180 degrees phase angle

الصفحة التالية

تنقل الاختبار